King Fahd University of Petroleum and Minerals

Department of Mathematics

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Math 102 Major Exam II Term 251

Original exam date: 12 November 2025. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

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1. The volume of the solid generated by revolving the region bounded by the graphs of $y = x^2$, $y = 4x - x^2$ about the x-axis is

(a) $\dfrac{34\pi}{3}$

(b) $\dfrac{32\pi}{3}$

(c) $\dfrac{37\pi}{3}$

(d) $\dfrac{31\pi}{3}$

(e) $\dfrac{35\pi}{3}$

2. $\int \frac{x^3}{4\sqrt{4+x^2}}\,dx =$

(a) $(x+8)\dfrac{\sqrt{4+x^2}}{12} + C$

(b) $(x-8)\dfrac{\sqrt{4+x^2}}{12} + C$

(c) $(x^2+8)\dfrac{\sqrt{4+x^2}}{12} + C$

(d) $(x^2-8)\dfrac{\sqrt{4+x^2}}{12} + C$

(e) $(x^2-8)\dfrac{\sqrt{4+x^2}}{6} + C$

3. $\int \frac{5\,dx}{x^2+3x-4} =$

(a) $\ln\left|\dfrac{x-1}{x+4}\right| + C$

(b) $\ln\left|\dfrac{x-1}{x-4}\right| + C$

(c) $\ln\left|\dfrac{x+1}{x-4}\right| + C$

(d) $\ln\left|\dfrac{x+1}{x+4}\right| + C$

(e) $\ln\left|\dfrac{x+1}{x-2}\right| + C$

4. If $\dfrac{9-x^2}{7x^3+x} = \dfrac{A}{x} + \dfrac{Bx+C}{7x^2+1}$, then $A+B+C =$

(a) -55

(b) -66

(c) -44

(d) -33

(e) -22

5. $\int \frac{4\,dx}{x^2\sqrt{16-x^2}} =$

(a) $\dfrac{-\sqrt{16-x^2}}{2x} + C$

(b) $\dfrac{-\sqrt{16-x^2}}{4x} + C$

(c) $\dfrac{\sqrt{16-x^2}}{2x} + C$

(d) $\dfrac{-\sqrt{16-x^2}}{x} + C$

(e) $\dfrac{\sqrt{16-x^2}}{4x} + C$

6. The arc length of the graph of the function $f(x) = \ln(\sin x)$ over the interval $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ is

(a) $\ln\left(\frac{2\sqrt{2}+1}{\sqrt{2}-1}\right)$

(b) $\ln\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$

(c) $\ln\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$

(d) $\ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

(e) $\ln\left(\frac{\sqrt{2}+1}{2\sqrt{2}-1}\right)$

7. The area of the surface generated by revolving the curve $y = \sqrt{9-x^2}$ on the interval $[-2, 2]$ about the x-axis is

(a) 26π

(b) 18π

(c) 22π

(d) 24π

(e) 20π

8. $\int \frac{xe^{2x}}{(2x+1)^2}\,dx =$

(a) $\frac{e^{2x}}{2x+1} + C$

(b) $\frac{e^{2x}}{4(2x+1)} + C$

(c) $\frac{e^{x}}{4(2x+1)} + C$

(d) $\frac{e^{x}}{2(2x+1)} + C$

(e) $\frac{e^{2x}}{2(2x+1)} + C$

9. $\int_0^1 x \arcsin x^2 \, dx =$

(a) $\frac{1}{4}(\pi - 2)$

(b) $\frac{1}{4}(\pi + 2)$

(c) $\frac{1}{2}(\pi + 2)$

(d) $\frac{1}{2}(\pi - 2)$

(e) $\frac{1}{8}(\pi - 2)$

10. $\int_0^{\pi/4} \tan^5 x \, dx =$

(a) $\frac{1 + \ln 2}{4}$

(b) $\frac{-1 + 2\ln 2}{4}$

(c) $\frac{1 - \ln 2}{4}$

(d) $\frac{1 - 2\ln 2}{4}$

(e) $\frac{1 + 2\ln 2}{4}$

11. $\int (\tan x)\,[\ln(\cos x)]\, dx =$

(a) $\frac{-[\ln(\cos x)]}{2} + C$

(b) $\frac{[\ln(\cos x)]^2}{2} + C$

(c) $\frac{-[\ln(\cos x)]^2}{2} + C$

(d) $[\ln(\cos x)]^2 + C$

(e) $-[\ln(\cos x)]^2 + C$

12. The volume of the solid formed by revolving the region bounded by the graphs of $y = \sqrt{x-1}$, $y = 0$, $x = 2$ about the y-axis is

(a) $\dfrac{38\pi}{15}$

(b) $\dfrac{37\pi}{15}$

(c) $\dfrac{31\pi}{15}$

(d) $\dfrac{32\pi}{15}$

(e) $\dfrac{34\pi}{15}$

13. $\int_0^{\pi/2} \frac{d\theta}{3-2\cos\theta} =$

(a) $\dfrac{2}{\sqrt{5}} \arctan\sqrt{5}$

(b) $\dfrac{5}{\sqrt{2}} \arctan\sqrt{5}$

(c) $\dfrac{4}{\sqrt{5}} \arctan\sqrt{5}$

(d) $\dfrac{3}{\sqrt{5}} \arctan\sqrt{5}$

(e) $\dfrac{1}{\sqrt{5}} \arctan\sqrt{5}$

14. $\int \frac{\cos^5 t}{\sqrt{\sin t}}\, dt =$

(a) $4\sqrt{\sin t} - \dfrac{4}{5}(\sin t)^{5/2} + \dfrac{2}{9}(\sin t)^{9/2} + C$

(b) $2\sqrt{\sin t} - \dfrac{4}{5}(\sin t)^{5/2} + \dfrac{2}{9}(\sin t)^{9/2} + C$

(c) $\sqrt{\sin t} - \dfrac{4}{5}(\sin t)^{5/2} + \dfrac{2}{9}(\sin t)^{9/2} + C$

(d) $2\sqrt{\sin t} - \dfrac{4}{5}(\sin t)^{5/2} - \dfrac{2}{9}(\sin t)^{9/2} + C$

(e) $2\sqrt{\sin t} + \dfrac{4}{5}(\sin t)^{5/2} + \dfrac{2}{9}(\sin t)^{9/2} + C$

Answer Key

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Q	Ans	Brief explanation
1	**(b)**	Washers on $[0,2]$: $\pi\int_0^2(16x^2-8x^3)\,dx=\frac{32\pi}{3}$.
2	**(d)**	$u=4+x^2$: $\frac{1}{8}\int(u^{1/2}-4u^{-1/2})du=\frac{\sqrt{u}(u-12)}{12}=(x^2-8)\frac{\sqrt{4+x^2}}{12}$.
3	**(a)**	$\frac{5}{(x-1)(x+4)}=\frac{1}{x-1}-\frac{1}{x+4}$, so $\ln\left\lvert\frac{x-1}{x+4}\right\rvert+C$.
4	**(a)**	$9-x^2=A(7x^2+1)+(Bx+C)x$: $A=9$, $C=0$, $B=-64$, sum -55.
5	**(b)**	$x=4\sin\theta$: $\frac{1}{4}\int\csc^2\theta\,d\theta=-\frac{1}{4}\cot\theta=-\frac{\sqrt{16-x^2}}{4x}$.
6	**(d)**	$L=\int\csc x\,dx=\ln\lvert\csc x-\cot x\rvert$ from $\frac{\pi}{4}$ to $\frac{3\pi}{4}$: $\ln\frac{\sqrt{2}+1}{\sqrt{2}-1}$.
7	**(d)**	$y\sqrt{1+y'^2}=3$, so $S=2\pi\int_{-2}^{2}3\,dx=24\pi$.
8	**(b)**	By parts with $v=-\frac{1}{2(2x+1)}$: $-\frac{xe^{2x}}{2(2x+1)}+\frac{e^{2x}}{4}=\frac{e^{2x}}{4(2x+1)}$.
9	**(a)**	$w=x^2$: $\frac{1}{2}\int_0^1\arcsin w\,dw=\frac{1}{2}(\frac{\pi}{2}-1)=\frac{1}{4}(\pi-2)$.
10	**(b)**	$\tan^5 x=\tan^3 x\sec^2 x-\tan x\sec^2 x+\tan x$ gives $-\frac{1}{4}+\frac{1}{2}\ln 2$.
11	**(c)**	$u=\ln\cos x$, $du=-\tan x\,dx$: $-\int u\,du=-\frac{[\ln(\cos x)]^2}{2}$.
12	**(d)**	Shells: $2\pi\int_1^2 x\sqrt{x-1}\,dx=2\pi\left(\frac{2}{5}+\frac{2}{3}\right)=\frac{32\pi}{15}$.
13	**(a)**	$u=\tan\frac{\theta}{2}$: $\int_0^1\frac{2\,du}{1+5u^2}=\frac{2}{\sqrt{5}}\arctan\sqrt{5}$.
14	**(b)**	$u=\sin t$: $\int(u^{-1/2}-2u^{3/2}+u^{7/2})du=2u^{1/2}-\frac{4}{5}u^{5/2}+\frac{2}{9}u^{9/2}$.